Scenario 7: nodes (page 1)

node	layer	class	end_use	service_weight	capacity_mw	capacity_m3_per_day	requirement	own_layer_response	failure_modes
SWM-FUE-0001	fuel	import_terminal	no				MW 10.3744; m3/day 8.93205	source	power: throttle; telecom: throttle; transport: throttle
SWM-FUE-0002	fuel	tank_farm	no				MW 1.27728; m3/day 10.5285	trip	power: trip
SWM-FUE-0003	fuel	depot	yes	4.0			MW 0.112303; m3/day 7.68856	trip	power: trip; telecom: throttle; transport: throttle
SWM-FUE-0004	fuel	depot	yes	4.0			MW 0.0635019; m3/day 11.9703	trip	power: trip; telecom: throttle; transport: throttle
SWM-HLT-0001	healthcare	tertiary	yes	10.0			MW 3.09536; m3/day 2803.7	source	power: step_to_floor (floor 0.6); telecom: trip; transport: throttle; water: trip
SWM-HLT-0002	healthcare	regional	yes	7.0			MW 1.74486; m3/day 252.821	source	power: step_to_floor (floor 0.6); telecom: throttle; transport: throttle; water: trip
SWM-HLT-0003	healthcare	clinic	yes	3.0			MW 0.0243536; m3/day 1.62625	source	power: trip; telecom: throttle; transport: trip; water: trip
SWM-HLT-0004	healthcare	blood_bank	yes	6.0			MW 0.0465964; m3/day 4.60035	source	power: trip; telecom: throttle; transport: throttle
SWM-POW-0001	power	generation	no		52.4		m3/day 1617.89; GJ/day 5834.56	source	fuel: trip; telecom: throttle; water: trip
SWM-POW-0002	power	transmission_substation	no					throttle	telecom: throttle
SWM-POW-0003	power	distribution_substation	yes	5.0				throttle	telecom: throttle
SWM-POW-0004	power	distribution_substation	yes	5.0				throttle	telecom: trip
SWM-TEL-0001	telecom	exchange	no				MW 0.187416	source	power: trip
SWM-TEL-0002	telecom	mast	yes	1.0			MW 0.00457391	trip	power: trip
SWM-TEL-0003	telecom	mast	yes	1.0			MW 0.00272203	trip	power: trip
SWM-TEL-0004	telecom	mast	yes	1.0			MW 0.00549639	trip	power: trip
SWM-TRN-0001	transport	seaport	yes	7.0			MW 6.92338; GJ/day 14367.4	source	fuel: throttle; power: trip; telecom: throttle
SWM-TRN-0002	transport	airport	yes	6.0			MW 0.196918; GJ/day 2691.38	source	fuel: trip; power: trip; telecom: trip
SWM-TRN-0003	transport	freight_yard	yes	4.0			MW 0.583885; GJ/day 93.1231	source	fuel: throttle; telecom: trip
SWM-TRN-0004	transport	bridge	no					source	
SWM-WAT-0001	water	reservoir	no					source	
SWM-WAT-0002	water	treatment	no			226300	MW 3.62111	throttle	power: throttle; telecom: throttle
SWM-WAT-0003	water	pumping	no			199800	MW 2.76	throttle	power: throttle; telecom: throttle
SWM-WAT-0004	water	storage	yes	5.0				trip	

Scenario 7: stocks (page 1)

stock	node	kind	covers	quantity	units	draw_per_h	hours	refill	generator_fuel	rule
SWM-FUE-0001:lng	SWM-FUE-0001	lng	transport	76895.87368290884	GJ	243.106726829482	316.305002	none	no	reserves.lng.days
SWM-FUE-0002:oil_government	SWM-FUE-0002	oil_government	fuel	1081672.1726156368	GJ	714.661749487883	1513.544237	none	no	reserves.oil.government_days
SWM-FUE-0002:oil_industry	SWM-FUE-0002	oil_industry	fuel	1565263.134054395	GJ	714.661749487883	2190.21535	none	no	reserves.oil.industry_days
SWM-HLT-0001:generator_fuel	SWM-HLT-0001	generator_fuel	power	41906.70923012689	L	499.5908164238084	83.882065	delivery	yes	reserves.hospital.generator_fuel_h
SWM-HLT-0001:supplies	SWM-HLT-0001	supplies	transport	114.22077179814225	hours of supplies	1.0	114.220772	none	no	reserves.hospital.supplies_h
SWM-HLT-0001:water	SWM-HLT-0001	water	water	1403.117547912844	m3	24.55	57.153464	main	no	reserves.hospital.water_h
SWM-HLT-0002:generator_fuel	SWM-HLT-0002	generator_fuel	power	15365.525275333406	L	281.61994913182707	54.56121	delivery	yes	reserves.hospital.generator_fuel_h
SWM-HLT-0002:supplies	SWM-HLT-0002	supplies	transport	33.11912581696437	hours of supplies	1.0	33.119126	none	no	reserves.hospital.supplies_h
SWM-HLT-0002:water	SWM-HLT-0002	water	water	548.7867701044333	m3	5.0874999999999995	107.869635	main	no	reserves.hospital.water_h
SWM-HLT-0004:blood	SWM-HLT-0004	blood	transport	127.27339667543612	hours of issue	1.0	127.273397	none	no	reserves.blood_bank.blood_stock_days
SWM-HLT-0004:generator_fuel	SWM-HLT-0004	generator_fuel	power	464.68756433935766	L	12.534425637686184	37.072904	delivery	yes	reserves.blood_bank.generator_fuel_h
SWM-POW-0001:purified_water	SWM-POW-0001	purified_water	water	687.4728540085878	m3	67.41207048677568	10.198068	main	no	reserves.generation.purified_water_h
SWM-TEL-0001:battery	SWM-TEL-0001	battery	power	448.4885900489903	kWh	187.41566278367821	2.393016	battery	no	reserves.exchange.battery_h
SWM-TEL-0001:generator_fuel	SWM-TEL-0001	generator_fuel	power	1329.935302397058	L	50.41481328880944	26.379852	delivery	yes	reserves.exchange.fuel_h
SWM-TEL-0002:battery	SWM-TEL-0002	battery	power	17.84456918763688	kWh	4.573906085451859	3.901385	battery	no	reserves.mast.battery_h
SWM-TEL-0003:battery	SWM-TEL-0003	battery	power	21.028980007759102	kWh	2.722028524425703	7.725481	battery	no	reserves.mast.battery_h
SWM-TEL-0004:battery	SWM-TEL-0004	battery	power	23.951322668125318	kWh	5.496388257254363	4.357648	battery	no	reserves.mast.battery_h
SWM-WAT-0002:generator_fuel	SWM-WAT-0002	generator_fuel	power	34476.8149422015	L	974.0793226044929	35.394258	delivery	yes	reserves.treatment.fuel_h
SWM-WAT-0004:distribution_water	SWM-WAT-0004	distribution_water	water	3975.260944353419	m3	165.56130698872812	24.010809	pumps	no	reserves.storage.peak_day_h

Scenario 7: generators (page 1)

node	fails_to_start	run_hours_to_failure
SWM-HLT-0001	no	836.4797456597048
SWM-HLT-0002	no	246.11607178695408
SWM-HLT-0004	no	259.94225795230903
SWM-TEL-0001	no	2094.769072055871
SWM-WAT-0002	no	3535.320098092566

Scenario 7: tolerance (page 1)

dependency	consumer	rule	T_h	ramp_to_h	d	kappa_public_share	d_eff	manual_switching
SWM-DEP-0017	SWM-POW-0001	tolerance.generation	12.0		0.95	0.2	0.99	no
SWM-DEP-0018	SWM-POW-0002	tolerance.substation	12.0		1.0	0.2	1.0	yes
SWM-DEP-0019	SWM-POW-0003	tolerance.substation	12.0		1.0	0.5	1.0	yes
SWM-DEP-0020	SWM-POW-0004	tolerance.substation	12.0		1.0	0.5	1.0	yes
SWM-DEP-0021	SWM-WAT-0002	tolerance.treatment	24.0		0.9	0.5	0.95	no
SWM-DEP-0022	SWM-WAT-0003	tolerance.pumping	12.0		0.8	0.5	0.9	no
SWM-DEP-0023	SWM-TRN-0001	tolerance.seaport_freight_yard	24.0		0.5	1.0	0.5	no
SWM-DEP-0024	SWM-TRN-0002	tolerance.towered_airport	0.0		0.2	1.0	0.19999999999999996	no
SWM-DEP-0025	SWM-TRN-0003	tolerance.seaport_freight_yard	24.0		0.5	1.0	0.5	no
SWM-DEP-0026	SWM-FUE-0001	tolerance.import_terminal	12.0		0.8	0.5	0.9	no
SWM-DEP-0027	SWM-FUE-0003	tolerance.depot	12.0		0.6	1.0	0.6	no
SWM-DEP-0028	SWM-FUE-0004	tolerance.depot	12.0		0.6	1.0	0.6	no
SWM-DEP-0029	SWM-HLT-0001	tolerance.hospital_records_on_site	24.0		0.85	1.0	0.85	no
SWM-DEP-0030	SWM-HLT-0002	tolerance.hospital_records_on_site	24.0		0.85	1.0	0.85	no
SWM-DEP-0031	SWM-HLT-0003	tolerance.clinic	24.0		0.8	1.0	0.8	no
SWM-DEP-0032	SWM-HLT-0004	tolerance.blood_bank	24.0		0.7	1.0	0.7	no

Scenario 7: dependencies (page 1)

edge	dependency	supplier	consumer	commodity	kappa	transfer_form	feed_configuration	feed_group	fuel_resupply	switched_on
SWM-CPL-0001	SWM-DEP-0001	SWM-POW-0003	SWM-TEL-0001	electricity	1.0	linear	single	1	no	True
SWM-CPL-0002	SWM-DEP-0002	SWM-POW-0002	SWM-TEL-0002	electricity	1.0	linear	two_half	1	no	True
SWM-CPL-0003	SWM-DEP-0002	SWM-POW-0004	SWM-TEL-0002	electricity	1.0	linear	two_half	1	no	True
SWM-CPL-0004	SWM-DEP-0003	SWM-POW-0002	SWM-TEL-0003	electricity	1.0	linear	single	1	no	True
SWM-CPL-0005	SWM-DEP-0004	SWM-POW-0004	SWM-TEL-0004	electricity	1.0	linear	two_full	1	no	True
SWM-CPL-0006	SWM-DEP-0004	SWM-POW-0002	SWM-TEL-0004	electricity	1.0	linear	two_full	2	no	True
SWM-CPL-0007	SWM-DEP-0005	SWM-POW-0002	SWM-WAT-0002	electricity	1.0	linear	single	1	no	True
SWM-CPL-0008	SWM-DEP-0006	SWM-POW-0003	SWM-WAT-0003	electricity	1.0	linear	two_full	1	no	True
SWM-CPL-0009	SWM-DEP-0006	SWM-POW-0002	SWM-WAT-0003	electricity	1.0	linear	two_full	2	no	True
SWM-CPL-0010	SWM-DEP-0007	SWM-POW-0002	SWM-TRN-0001	electricity	1.0	linear	single	1	no	False
SWM-CPL-0011	SWM-DEP-0008	SWM-POW-0002	SWM-TRN-0002	electricity	1.0	linear	two_full	1	no	False
SWM-CPL-0012	SWM-DEP-0008	SWM-POW-0003	SWM-TRN-0002	electricity	1.0	linear	two_full	2	no	False
SWM-CPL-0013	SWM-DEP-0009	SWM-POW-0002	SWM-FUE-0001	electricity	1.0	linear	single	1	no	False
SWM-CPL-0014	SWM-DEP-0010	SWM-POW-0003	SWM-FUE-0002	electricity	1.0	linear	two_full	1	no	False
SWM-CPL-0015	SWM-DEP-0010	SWM-POW-0002	SWM-FUE-0002	electricity	1.0	linear	two_full	2	no	False
SWM-CPL-0016	SWM-DEP-0011	SWM-POW-0002	SWM-FUE-0003	electricity	1.0	linear	two_half	1	no	False
SWM-CPL-0017	SWM-DEP-0011	SWM-POW-0003	SWM-FUE-0003	electricity	1.0	linear	two_half	1	no	False
SWM-CPL-0018	SWM-DEP-0012	SWM-POW-0004	SWM-FUE-0004	electricity	1.0	linear	single	1	no	False
SWM-CPL-0019	SWM-DEP-0013	SWM-POW-0003	SWM-HLT-0001	electricity	1.0	linear	single	1	no	True
SWM-CPL-0020	SWM-DEP-0014	SWM-POW-0002	SWM-HLT-0002	electricity	1.0	linear	two_full	1	no	True
SWM-CPL-0021	SWM-DEP-0014	SWM-POW-0004	SWM-HLT-0002	electricity	1.0	linear	two_full	2	no	True
SWM-CPL-0022	SWM-DEP-0015	SWM-POW-0002	SWM-HLT-0003	electricity	1.0	linear	single	1	no	True
SWM-CPL-0023	SWM-DEP-0016	SWM-POW-0003	SWM-HLT-0004	electricity	1.0	linear	single	1	no	True
SWM-CPL-0024	SWM-DEP-0017	SWM-TEL-0001	SWM-POW-0001	communications	0.2	threshold	single	1	no	True
SWM-CPL-0025	SWM-DEP-0018	SWM-TEL-0001	SWM-POW-0002	communications	0.2	threshold	single	1	no	True
SWM-CPL-0026	SWM-DEP-0019	SWM-TEL-0003	SWM-POW-0003	communications	0.5	threshold	two_full	1	no	True
SWM-CPL-0027	SWM-DEP-0019	SWM-TEL-0004	SWM-POW-0003	communications	0.5	threshold	two_full	2	no	True
SWM-CPL-0028	SWM-DEP-0020	SWM-TEL-0001	SWM-POW-0004	communications	0.5	threshold	two_half	1	no	True
SWM-CPL-0029	SWM-DEP-0020	SWM-TEL-0003	SWM-POW-0004	communications	0.5	threshold	two_half	1	no	True
SWM-CPL-0030	SWM-DEP-0021	SWM-TEL-0001	SWM-WAT-0002	communications	0.5	threshold	two_full	1	no	False
SWM-CPL-0031	SWM-DEP-0021	SWM-TEL-0004	SWM-WAT-0002	communications	0.5	threshold	two_full	2	no	False
SWM-CPL-0032	SWM-DEP-0022	SWM-TEL-0001	SWM-WAT-0003	communications	0.5	threshold	single	1	no	False
SWM-CPL-0033	SWM-DEP-0023	SWM-TEL-0001	SWM-TRN-0001	communications	1.0	linear	single	1	no	False
SWM-CPL-0034	SWM-DEP-0024	SWM-TEL-0001	SWM-TRN-0002	communications	1.0	threshold	single	1	no	False
SWM-CPL-0035	SWM-DEP-0025	SWM-TEL-0002	SWM-TRN-0003	communications	1.0	linear	two_full	1	no	False
SWM-CPL-0036	SWM-DEP-0025	SWM-TEL-0001	SWM-TRN-0003	communications	1.0	linear	two_full	2	no	False
SWM-CPL-0037	SWM-DEP-0026	SWM-TEL-0003	SWM-FUE-0001	communications	0.5	threshold	two_full	1	no	False
SWM-CPL-0038	SWM-DEP-0026	SWM-TEL-0001	SWM-FUE-0001	communications	0.5	threshold	two_full	2	no	False
SWM-CPL-0039	SWM-DEP-0027	SWM-TEL-0001	SWM-FUE-0003	communications	1.0	linear	single	1	no	False
SWM-CPL-0040	SWM-DEP-0028	SWM-TEL-0004	SWM-FUE-0004	communications	1.0	linear	single	1	no	False
SWM-CPL-0041	SWM-DEP-0029	SWM-TEL-0001	SWM-HLT-0001	communications	1.0	linear	single	1	no	False
SWM-CPL-0042	SWM-DEP-0030	SWM-TEL-0004	SWM-HLT-0002	communications	1.0	linear	two_half	1	no	False
SWM-CPL-0043	SWM-DEP-0030	SWM-TEL-0003	SWM-HLT-0002	communications	1.0	linear	two_half	1	no	False
SWM-CPL-0044	SWM-DEP-0031	SWM-TEL-0001	SWM-HLT-0003	communications	1.0	linear	two_full	1	no	False
SWM-CPL-0045	SWM-DEP-0031	SWM-TEL-0003	SWM-HLT-0003	communications	1.0	linear	two_full	2	no	False
SWM-CPL-0046	SWM-DEP-0032	SWM-TEL-0001	SWM-HLT-0004	communications	1.0	linear	single	1	no	False
SWM-CPL-0047	SWM-DEP-0033	SWM-WAT-0003	SWM-POW-0001	water	1.0	linear	single	1	no	False
SWM-CPL-0048	SWM-DEP-0034	SWM-WAT-0004	SWM-HLT-0001	water	1.0	linear	single	1	no	True
SWM-CPL-0049	SWM-DEP-0035	SWM-WAT-0004	SWM-HLT-0002	water	1.0	linear	two_full	1	no	True
SWM-CPL-0050	SWM-DEP-0035	SWM-WAT-0003	SWM-HLT-0002	water	1.0	linear	two_full	2	no	True
SWM-CPL-0051	SWM-DEP-0036	SWM-WAT-0003	SWM-HLT-0003	water	1.0	linear	single	1	no	True
SWM-CPL-0052	SWM-DEP-0037	SWM-TRN-0001	SWM-FUE-0001	transport access	1.0	threshold	any_reachable	1	no	True
SWM-CPL-0053	SWM-DEP-0038	SWM-TRN-0003	SWM-FUE-0003	transport access	1.0	threshold	any_reachable	1	no	True
SWM-CPL-0054	SWM-DEP-0039	SWM-TRN-0003	SWM-FUE-0004	transport access	1.0	threshold	any_reachable	1	no	True
SWM-CPL-0055	SWM-DEP-0040	SWM-TRN-0002	SWM-HLT-0001	transport access	1.0	threshold	any_reachable	1	no	False
SWM-CPL-0056	SWM-DEP-0040	SWM-TRN-0001	SWM-HLT-0001	transport access	1.0	threshold	any_reachable	2	no	False
SWM-CPL-0057	SWM-DEP-0040	SWM-TRN-0003	SWM-HLT-0001	transport access	1.0	threshold	any_reachable	3	no	False

Scenario 7: dependencies (page 2)

SWM-CPL-0058	SWM-DEP-0041	SWM-TRN-0003	SWM-HLT-0002	transport access	1.0	threshold	any_reachable	1	no	False
SWM-CPL-0059	SWM-DEP-0041	SWM-TRN-0001	SWM-HLT-0002	transport access	1.0	threshold	any_reachable	2	no	False
SWM-CPL-0060	SWM-DEP-0041	SWM-TRN-0002	SWM-HLT-0002	transport access	1.0	threshold	any_reachable	3	no	False
SWM-CPL-0061	SWM-DEP-0042	SWM-TRN-0003	SWM-HLT-0003	transport access	1.0	threshold	any_reachable	1	no	False
SWM-CPL-0062	SWM-DEP-0043	SWM-TRN-0002	SWM-HLT-0004	transport access	1.0	threshold	any_reachable	1	no	False
SWM-CPL-0063	SWM-DEP-0043	SWM-TRN-0003	SWM-HLT-0004	transport access	1.0	threshold	any_reachable	2	no	False
SWM-CPL-0064	SWM-DEP-0044	SWM-FUE-0001	SWM-POW-0001	fuel	1.0	linear	single	1	no	True
SWM-CPL-0065	SWM-DEP-0045	SWM-FUE-0002	SWM-TEL-0001	fuel	1.0	threshold	single	1	yes	False
SWM-CPL-0066	SWM-DEP-0049	SWM-FUE-0003	SWM-WAT-0002	fuel	1.0	threshold	single	1	yes	False
SWM-CPL-0067	SWM-DEP-0051	SWM-FUE-0003	SWM-TRN-0001	fuel	1.0	threshold	two_full	1	no	False
SWM-CPL-0068	SWM-DEP-0051	SWM-FUE-0002	SWM-TRN-0001	fuel	1.0	threshold	two_full	2	no	False
SWM-CPL-0069	SWM-DEP-0052	SWM-FUE-0002	SWM-TRN-0002	fuel	1.0	linear	single	1	no	False
SWM-CPL-0070	SWM-DEP-0053	SWM-FUE-0002	SWM-TRN-0003	fuel	1.0	threshold	single	1	no	False
SWM-CPL-0071	SWM-DEP-0054	SWM-FUE-0003	SWM-HLT-0001	fuel	1.0	threshold	single	1	yes	False
SWM-CPL-0072	SWM-DEP-0055	SWM-FUE-0004	SWM-HLT-0002	fuel	1.0	threshold	single	1	yes	False
SWM-CPL-0073	SWM-DEP-0057	SWM-FUE-0004	SWM-HLT-0004	fuel	1.0	threshold	single	1	yes	False

Scenario 7: links (page 1)

A node fed by more than one link of its own layer takes the largest upstream output (capped at 1); a link on a failed line passes nothing until repaired.
A commodity with more than one feed group: each group delivers the smallest κ x g among its edges, the groups are summed, capped at 1 (telecom edges deliver g alone). A node's state is its condition x the lowest response across its commodities.

link	layer	from	to	length_km	on_towers	commodity	flow
SWM-FUE-L0001	fuel	SWM-FUE-0001	SWM-FUE-0002	1.389142	no	fuel	SWM-FUE-0001 -> SWM-FUE-0002
SWM-FUE-L0002	fuel	SWM-FUE-0002	SWM-FUE-0003	2.145458	no	fuel	SWM-FUE-0002 -> SWM-FUE-0003
SWM-FUE-L0003	fuel	SWM-FUE-0002	SWM-FUE-0004	13.889509	no	fuel	SWM-FUE-0002 -> SWM-FUE-0004
SWM-HLT-L0001	healthcare	SWM-HLT-0001	SWM-HLT-0002	12.230995	no	none (patient referral)	none
SWM-HLT-L0002	healthcare	SWM-HLT-0002	SWM-HLT-0003	9.878433	no	none (patient referral)	none
SWM-HLT-L0003	healthcare	SWM-HLT-0001	SWM-HLT-0004	1.618692	no	none (patient referral)	none
SWM-POW-L0001	power	SWM-POW-0001	SWM-POW-0002	7.878653	yes	electricity	SWM-POW-0001 -> SWM-POW-0002
SWM-POW-L0002	power	SWM-POW-0002	SWM-POW-0003	8.874613	yes	electricity	SWM-POW-0002 -> SWM-POW-0003
SWM-POW-L0003	power	SWM-POW-0002	SWM-POW-0004	6.690608	yes	electricity	SWM-POW-0002 -> SWM-POW-0004
SWM-POW-L0004	power	SWM-POW-0001	SWM-POW-0003	2.780797	yes	electricity	SWM-POW-0001 -> SWM-POW-0003
SWM-POW-L0005	power	SWM-POW-0001	SWM-POW-0004	14.096069	yes	electricity	SWM-POW-0001 -> SWM-POW-0004
SWM-TEL-L0001	telecom	SWM-TEL-0001	SWM-TEL-0002	20.910496	no	telecom service	SWM-TEL-0001 -> SWM-TEL-0002
SWM-TEL-L0002	telecom	SWM-TEL-0001	SWM-TEL-0003	15.649747	no	telecom service	SWM-TEL-0001 -> SWM-TEL-0003
SWM-TEL-L0003	telecom	SWM-TEL-0001	SWM-TEL-0004	12.552448	no	telecom service	SWM-TEL-0001 -> SWM-TEL-0004
SWM-TRN-L0001	transport	SWM-TRN-0004	SWM-TRN-0002	12.668271	no	none (roads: crew travel only)	none
SWM-TRN-L0002	transport	SWM-TRN-0004	SWM-TRN-0003	6.652784	no	none (roads: crew travel only)	none
SWM-TRN-L0003	transport	SWM-TRN-0001	SWM-TRN-0002	3.8835	no	none (roads: crew travel only)	none
SWM-TRN-L0004	transport	SWM-TRN-0001	SWM-TRN-0003	4.541469	no	none (roads: crew travel only)	none
SWM-TRN-L0005	transport	SWM-TRN-0002	SWM-TRN-0003	6.017001	no	none (roads: crew travel only)	none
SWM-WAT-L0001	water	SWM-WAT-0001	SWM-WAT-0002	1.129015	no	water	SWM-WAT-0001 -> SWM-WAT-0002
SWM-WAT-L0002	water	SWM-WAT-0002	SWM-WAT-0003	4.908655	no	water	SWM-WAT-0002 -> SWM-WAT-0003
SWM-WAT-L0003	water	SWM-WAT-0003	SWM-WAT-0004	3.204405	no	water	SWM-WAT-0003 -> SWM-WAT-0004

Scenario 7: damage_lines (page 1)

line	structure	peak_gust_m_per_s	failure_point	km_along_line	road_km	offroad_km	by_sea	travel_h_from_base	crew_work_h
SWM-POW-L0004	tower	83.594	1	1.3905	1.013322	3.447662	no	1.749164	72.0
SWM-TEL-L0001	pole	95.01	1	12.5	5.155214	8.749773	no	4.503767	6.0
SWM-TEL-L0002	pole	83.668	1	9.5	5.447586	5.724775	no	2.998577	6.0
SWM-TEL-L0002	pole	83.668	2	11.5	5.623446	7.717028	no	3.9991	6.0

Scenario 7: damage_facilities (page 1)

node damage_state flood_depth_m repairable crew_work_h parts_lead_h
(none)

Scenario 7: crews (page 1)

crew	base	base_km_east	base_km_north
CRW-0001	CRB-0001	22.685994	20.024881
CRW-0002	CRB-0001	22.685994	20.024881

Scenario 7: travel (page 1)

node	road_km	offroad_km	by_sea	travel_h_from_base
SWM-FUE-0001	3.33704	0.0	no	0.083426
SWM-FUE-0002	5.032502	0.0	no	0.125813
SWM-FUE-0003	3.367161	0.0	no	0.084179
SWM-FUE-0004	15.198661	0.0	no	0.379967
SWM-HLT-0001	3.882063	0.0	no	0.097052
SWM-HLT-0002	15.233275	0.0	no	0.380832
SWM-HLT-0003	11.863007	0.0	no	0.296575
SWM-HLT-0004	2.761586	0.0	no	0.06904
SWM-POW-0001	3.162902	0.0	no	0.079073
SWM-POW-0002	9.042125	0.0	no	0.226053
SWM-POW-0003	5.641671	0.0	no	0.141042
SWM-POW-0004	13.525074	0.0	no	0.338127
SWM-TEL-0001	1.373789	0.0	no	0.034345
SWM-TEL-0002	29.178439	0.0	no	0.729461
SWM-TEL-0003	23.317304	0.0	no	0.582933
SWM-TEL-0004	16.117422	0.0	no	0.402936
SWM-TRN-0001	4.896822	0.0	no	0.122421
SWM-TRN-0002	1.013322	0.0	no	0.025333
SWM-TRN-0003	7.030323	0.0	no	0.175758
SWM-TRN-0004	13.681593	0.0	no	0.34204
SWM-WAT-0001	12.801172	0.0	no	0.320029
SWM-WAT-0002	10.771495	0.0	no	0.269287
SWM-WAT-0003	7.74255	0.0	no	0.193564
SWM-WAT-0004	7.680571	0.0	no	0.192014

Scenario 7: crew_order (page 1)

A free crew takes the open job with the smallest key (group, then -Lifeline weight), then the shortest trip from where it stands, then the lowest job id. It travels by road, then off road to a line; it never returns to base between jobs.

job	group	lifeline_weight	why
repair:SWM-POW-L0004	0		towered line
repair:SWM-TEL-L0001	2		everything else
repair:SWM-TEL-L0002	2		everything else

Scenario 7: lifeline_weights (page 1)

class	lifeline_weight
tertiary	10.0
regional	7.0
seaport	7.0
blood_bank	6.0
airport	6.0
distribution_substation	5.0
storage	5.0
depot	4.0
freight_yard	4.0
clinic	3.0
mast	1.0